

1. Administrative & Study Identification

Full Study Title: BRISOTE: A Multicentre, Randomised, Double-Blind, Parallel Group, Active-Controlled, Phase 3b Study to Evaluate the Efficacy and Safety of Benralizumab 30 mg SC in Eosinophilic Asthma Patients Uncontrolled on Medium-Dose Inhaled Corticosteroid Plus Long-acting β 2-Agonist **Short Title/Acronym:** BRISOTE **Protocol Number:** D3250C00101 **NCT Number:** NCT06750289 **Sponsor Name:** AstraZeneca **Investigational Product:** Benralizumab (30 mg SC) **Phase of Development:** Phase 3b **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of benralizumab as an add-on therapy in uncontrolled eosinophilic asthma participants treated with medium-dose ICS-LABA compared to the conventional treatment step of escalation of inhaled therapy to high-dose ICS-LABA. **Secondary Objectives:** To evaluate the reduction in annualized asthma exacerbation rate (AAER), improvement in lung function (FEV1), asthma symptom control, Health-Related Quality of Life (HRQoL), and overall safety and tolerability.

2.2 Background & Rationale To address the ongoing burden of disease in patients with uncontrolled eosinophilic asthma despite standard medium-dose inhaled corticosteroid and long-acting β 2-agonist (ICS-LABA) treatment. The study investigates whether introducing targeted biologic therapy (benralizumab, an anti-IL-5R α monoclonal antibody) earlier in the treatment pathway provides superior asthma control and exacerbation reduction compared to the conventional guideline recommendation of escalating to high-dose ICS-LABA, thereby potentially sparing patients from higher steroid exposure.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Primary Purpose:** Treatment **Control Type:** Active-Controlled (Escalation to high-dose ICS-LABA + placebo SC) **Blinding:** Double-blind **Randomization:** 1:1 Parallel Assignment **Allocation:** Randomized

3.2 Planned Enrollment & Duration Target **\$N\$:** 400 evaluable participants **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 28-Mar-2025 **Estimated Primary Completion:** 03-Nov-2027 **Estimated Study Completion:** 03-Nov-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 12 to 75 years. 2. Sex: All. 3. Documented history of physician-diagnosed asthma requiring treatment with at least medium-dose ICS (> 250 µg fluticasone dry powder formulation equivalents total daily dose) and a LABA for at least 12 months prior to Visit 1, and at least 2 asthma exacerbations in the prior 12 months. 4. Peripheral blood eosinophil count of ≥ 150 cells/ μ L.

4.2 Exclusion Criteria 1. Healthy volunteers are not accepted. 2. Important pulmonary disease other than asthma (or disease linked to high eosinophil counts). 3. Current smokers or former smokers with a smoking history ≥ 10 pack-years (former smokers must have stopped for at least 6 months prior to Visit 1). 4. History of alcohol or drug abuse within 12 months prior to the date informed consent is obtained.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Benralizumab 30 mg or placebo administered every 4 weeks for the first 3 doses, then every 8 weeks (up to 7 total injections). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 48 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Medium-dose ICS-LABA (and standard required asthma rescue medications). **Prohibited:** Maintenance oral corticosteroids (OCS) and concurrent participation in another clinical study with an investigational product.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to Day 0	Informed Consent, Medical History, Eosinophil Labs, e-Diary Run-in ($\geq 70\%$ compliance check)
Baseline	Day 0	Randomization, First SC Dose, ACQ-6 & Spirometry
Interim	Weeks 4, 8, 16, 24, 32	SC Injections (Q4W for 3 doses, then Q8W), Safety Labs, Efficacy Assessments (FEV1, ACQ-6)
End of Study	Week 48	Final SC Injection Follow-up, Final Vital Signs, Efficacy Assessment, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Annualized Asthma Exacerbation Rate (AAER) over the 48-week treatment period. **Measurement**

Method: Clinical assessment of documented asthma exacerbations requiring systemic corticosteroids, emergency room visits, or hospitalizations.

7.2 Secondary & Exploratory Endpoints 1. Change from baseline in pre-bronchodilator forced expiratory volume in 1 second (FEV1). 2. Change from baseline in Asthma Control Questionnaire (ACQ-6) score and Health-Related Quality of Life (HRQoL) metrics. 3. Time to first asthma exacerbation.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine evaluation of AEs via patient reporting, physician interviews, and specific monitoring for injection site reactions and hypersensitivity. **Serious Adverse Events (SAEs):** Required reporting timeline is typically within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent external DMC monitoring unblinded safety data throughout the trial phase.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary and secondary efficacy evaluations; Safety Set for all participants who receive at least one dose of the investigational product. **Sample Size Justification:** $N = 400$ powered ($\alpha = 0.05$, $\text{Power} \geq 80\%$) to demonstrate statistical superiority of the benralizumab add-on arm versus the active-comparator escalation arm in reducing the annualized asthma exacerbation rate. **Handling of Missing Data:** Multiple Imputation (MI) and Mixed-effect Model Repeated Measure (MMRM) for continuous efficacy variables (e.g., FEV1, ACQ-6).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Risk-based monitoring incorporating both onsite visits and remote data verification. **Audit Trail:** Robust electronic Case Report Form (eCRF) tracking, e-diary data validation, and regular site audits by AstraZeneca clinical quality assurance teams.

11. Study Identifiers & Governance

Primary ID: D3250C00101 **Registry Number:** NCT06750289 **CTIS Identifier:** 2024-515162-13-00 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** BRISOTE **Official Regulatory Title:**

BRISOTE: A Multicentre, Randomised, Double-Blind, Parallel Group, Active-Controlled, Phase 3b Study to Evaluate the Efficacy and Safety of Benralizumab 30 mg SC in Eosinophilic Asthma Patients Uncontrolled on Medium-Dose Inhaled Corticosteroid Plus Long-acting β 2-Agonist

12. Clinical Framework (Asthma Specific)

Primary Condition: Uncontrolled Eosinophilic Asthma **Diagnosis Threshold:** Blood eosinophils ≥ 150 cells/ μ L and ≥ 2 exacerbations in the past 12 months **Medical Methodology:** Add-on targeted biologic therapy (anti-IL-5R α) vs. Standard of Care step-up (High-Dose ICS-LABA) **Optimization Target:** Significant reduction in asthma exacerbation rate and improved symptom control without escalation of inhaled corticosteroids.

13. Study Procedures & Methodology

Management Pathway: Comparison of biologic add-on to conventional GINA escalation therapy. **Education Protocol (HCP):** Protocol training, accurate ACQ-6 administration, and SC injection procedures. **Education Protocol (Patient):** Daily asthma controller compliance (e-diary) tracking and exacerbation symptom recognition. **Quality Audit Mechanism:** Electronic diary adherence monitoring (requires $\geq 70\%$ run-in compliance) and clinical site source data verification.

14. Observation & Follow-Up Schedule

Total Duration: 48 weeks **Onsite Visit Cadence:** 7 visits (Screening, Baseline, Weeks 4, 8, 16, 24, 32, 48) **Remote Monitoring Frequency:** Intermittent e-diary compliance tracking between injection visits. **Checklist Review Interval:** Asthma daily diary reviewed at every scheduled onsite visit.

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator			[Insert Name]			_____		[DD-MMM-YYYY]								
	Clinical Operations Lead			[Insert Name]			_____		[DD-MMM-YYYY]								
	Lead Statistician			[Insert Name]			_____		[DD-MMM-YYYY]								